SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Error Analysis Task

Course Code:	CSA17	**Course Title:**	Artificial Intelligence
CO Assessed:	CO1 – Apply (BL3)	**Assessment:**	Assessment Tool 3 – Error Analysis Task
Weightage:	35% of CIA	**Duration / Marks:**	60 mins / 100 marks
CO1 Statement:	Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)		
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Section / Batch:	Batch 1	**Date:**	08.10.2026

Code of Conduct
I, Prabhakaran, 1912260199 (Name / Reg. No.) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the Student: _____

Instructions

- Read each question carefully; answer all four questions (100 in total). Off-Class Assessment Tool.
- Each question gives a student's faulty work. Do not solve from scratch: identify every error (cite it by statement, component, iteration or step number), explain its root cause, correct it with justification and apply the corrected concept. An error merely carried forward counts once.
- Follow the conventions stated in each question (alphabetical expansion, explored set, goal test on removal from the frontier, tie-breaking) and present tables neatly.
- Marks follow the Error Analysis rubric: error identification, root cause analysis, correction and justification, application of concepts, and clarity and presentation.
- Upload in GitHub and GCR (question paper with Reg. No., Name, answers, rubrics and evaluation page at the end).

Section / Topic	Focus	Question Nos.
1. Foundations and History of AI	Four approaches to AI, Turing Test, eras and milestones	**Q1**
2. Intelligent Agents: Agents and Environments	Agent, percepts, actions, agent function and program, PEAS	**Q1**
3. The Concept of Rationality	Performance measure, percept sequence, rational vs omniscient	**Q1**
4. The Nature of Environments	Task-environment properties and their classification	**Q1**
5. The Structure of Agents	Simple reflex and model-based reflex agent programs	**Q1**
6. Problem-Solving Agents	Goal formulation, problem formulation, search, execution	**Q2**
7. Problem Formulation	States, initial state, actions, goal test, path cost; belief states	**Q2, Q4**
8. Searching for Solutions	Search graph, frontier, explored set, node expansion	**Q3**

9. Uninformed Search Strategies	Error tracing of BFS, DFS and uniform-cost search	**Q3**
10. Search with Partial Information	Sensorless problems, belief states, percept update	**Q4**

Annexure A – Error Analysis Task

Question 1: Foundations of AI, Intelligent Agents and Rationality – Error Analysis of Concept Notes

Topic: Foundations and history of AI, agents and environments, rationality, nature of environments, structure of agents

A junior engineer prepared the revision notes below on the foundations of AI and intelligent agents. The notes contain errors; some statements may be correct. Analyse them statement by statement.

No.	Statement in the notes
S1	The Turing Test (1950) judges a machine to be intelligent if its internal reasoning always follows the laws of logic. It is therefore the standard example of the "thinking rationally" approach to AI.
S2	The "AI winters", periods of reduced funding and interest, occurred in the 1950s, before the 1956 Dartmouth workshop. That workshop then revived the field and coined the term "artificial intelligence".
S3	A rational agent always achieves the best possible outcome because it knows the actual result of every action in advance. Hence a taxi that picks a route later blocked by an unforeseeable landslide was not rational.
S4	PEAS description of an automated taxi – Performance measure: steering, accelerator, brake, horn. Environment: roads, other traffic, pedestrians, weather. Actuators: safe, fast, legal and comfortable trip. Sensors: cameras, GPS, speedometer.
S5	Chess played without a clock is a partially observable, stochastic, episodic, static and discrete task environment.
S6	A model-based reflex agent keeps an internal state, updated using knowledge of how the world evolves and how its own actions affect it. It can therefore act sensibly even when the current percept does not reveal the full state of the world.

(a) Identify which statements are correct and which are incorrect. For each incorrect statement, quote the erroneous phrase or claim (one statement contains more than one error).

Statement	Correct / Incorrect	Erroneous phrase / claim
S1	Incorrect	"internal reasoning always follows the laws of logic" and "thinking rationally". Turing Test tests human-like conversational behaviour.
S2	Incorrect	AI winters did not occur before Dartmouth in the 1950s; they occurred later, notably in the 1970s and late 1980s/early 1990s.
S3	Incorrect	"knows the actual result…in advance" and "was not rational". Rationality uses available percepts, knowledge, actions and performance measure; it does not require foresight.
S4	Incorrect	PEAS labels are swapped: steering/brake etc. are actuators, while safe/fast/legal/comfortable are performance measures.
S5	Incorrect	Chess without a clock is fully observable, deterministic, sequential and static; it is discrete. Not partially observable, stochastic or episodic.
S6	Correct	No erroneous phrase. This correctly describes a model-based reflex agent.

(b) Root cause analysis: for every error, explain the underlying misconception, i.e. which concept has been misunderstood and why the statement goes wrong.

Error (statement / part)	Root cause (misconception)

S1	Confuses the Turing Test (acting humanly) with the laws-of-thought / thinking-rationally approach.
S2	Misplaces AI winters chronologically and wrongly suggests the field had to be revived before its naming at Dartmouth in 1956.
S3	Confuses rationality with omniscience; rational actions are judged using information available at decision time.
S4	Does not distinguish PEAS categories: performance measure describes success; actuators perform actions.
S5	Misclassifies chess properties: board/state is visible, rules determine outcomes, moves affect later states, and no-clock chess is static.
S3/S4	Assumes certainty or confuses outcome measures with action mechanisms.

(c) Rewrite each faulty statement correctly and justify your correction. For a statement that is already correct, write "No change" and justify why.

Statement	Corrected statement	Justification
S1	The Turing Test evaluates whether a machine's behaviour in conversation is indistinguishable from a human's. It is an example of the acting-humanly approach.	It tests observable behaviour, not whether internal reasoning always follows formal logic.
S2	The Dartmouth workshop in 1956 helped establish AI as a field and introduced the term "artificial intelligence". AI winters were later periods of reduced funding and interest.	The chronology is corrected; winters followed earlier growth and disappointment.
S3	A rational agent selects an action expected to maximize its performance measure based on the percept sequence and available knowledge. An unforeseeable landslide does not by itself make the earlier route choice irrational.	Rationality depends on information available before acting, not perfect prediction of every outcome.
S4	Performance measure: safety, legality, travel time, comfort. Environment: roads, traffic, pedestrians, weather. Actuators: steering, accelerator, brakes, horn. Sensors: cameras, GPS, speedometer.	Each PEAS component is placed in its correct category.
S5	No change to the topic of chess being discrete and static without a clock, but correct the statement to: fully observable, deterministic, sequential, static and discrete.	The board and pieces are visible; outcomes follow fixed rules; moves affect later moves; without a clock the world does not change while deliberating; states/actions are discrete.
S6	No change.	The internal state uses a model of world evolution and action effects to interpret incomplete current percepts.

(d) Apply the corrected concepts to the following case. A self-driving taxi chooses a highway route that every available traffic report showed to be clear; an unforeseeable landslide then blocks the road.

(i) Using the four factors that decide whether an agent is rational, argue whether the taxi's choice was rational.

(ii) Name the properties of the taxi's task environment that make perfect foresight impossible.

(iii) Describe one situation in which a simple reflex taxi would behave poorly, and state what a model-based reflex taxi keeps in order to avoid it.

(i)	

(ii)	(i) Yes, rational. The action maximized expected performance based on the percept sequence, prior knowledge, available actions and performance measure. Reports showed the highway clear; the landslide was unforeseeable. A bad outcome alone does not prove irrationality.
(iii)	(iii) A simple reflex taxi may continue because the current view looks clear, ignoring a previously detected closure. A model-based reflex taxi keeps an internal state/model of road conditions, world changes and action effects.

Question 2: Problem-Solving Agents and Problem Formulation – Error Analysis of a Formulation

Topic: Problem-solving agents, problem formulation

A student formulates the **Missionaries and Cannibals** problem for a problem-solving agent. Three missionaries and three cannibals stand on the left bank of a river, with a boat that carries one or two people and cannot cross empty. The boat is on the left bank at the start. Missionaries must never be outnumbered by cannibals on either bank (a bank with no missionaries is safe). The objective is to move everyone to the right bank using the **fewest crossings**.

Component	Student's formulation
State representation	(M, C) = number of missionaries and cannibals on the left bank
Initial state	(3, 3)
Actions	Take the boat across with 1M, 2M, 1C, 2C or 1M + 1C
Goal test	M = C
Legality check	A state is legal if C ≤ M on the left bank (or M = 0 there)
Path cost	Number of people carried in the crossing
Note N1	The agent works in the order: search → goal formulation → problem formulation → execution.
Note N2	The state space contains 16 states, of which 10 are legal.

(a) For each component and note, state whether it is correct or incorrect; for each incorrect one, state what is wrong.

Component / Note	Correct / Incorrect	What is wrong
State representation	Incorrect	State must include boat side: (M, C, boat side).
Initial state	Correct	(3,3,L) is the initial state; the student's pair omits boat side.
Actions	Incorrect	The action list is incomplete: legal boat loads also include one missionary + one cannibal (already listed) and it must specify crossings in either direction; legality must be checked after each move.
Goal test	Incorrect	Goal is M=0 and C=0 on the left bank, with boat on the right; M=C is not the goal.
Legality check	Incorrect	Must check both banks: left is safe if M=0 or M≥C; right is safe if 3−M=0 or 3−M≥3−C.
Path cost	Incorrect	Each crossing costs 1, because objective is fewest crossings, not number of passengers.
Note N1	Incorrect	Correct order is goal formulation → problem formulation → search → execution.
Note N2	Partly incorrect	There are 16 (M,C) pairs and 10 satisfy both-bank safety, but full states include boat side: 32 combinations, 20 legal full states.

(b) Root cause analysis: for every error, explain the underlying cause, i.e. which principle of problem formulation or of the problem-solving agent has been violated.

Error (component / note)	Root cause
State representation	Boat location omitted; the state must contain all information needed to determine legal actions and future states.
Goal test	Equality M=C does not mean everyone crossed; goal requires no missionaries or cannibals left.
Legality check	Checks only the left bank and can allow cannibals to outnumber missionaries on the right.
Path cost	Optimizes passengers transported rather than the stated objective of minimizing crossings.
Note N1	The agent must define the goal and formalize the problem before searching and executing.
Note N2	Counts only (M,C) pairs, omitting boat side from the full state representation.

(c) Give the corrected version of every faulty component or note and justify each correction.

Component / Note	Corrected formulation	Justification
State representation	(M,C,B), where M,C are counts on the left and B {L,R}.	Boat side must be included. There are 16 (M,C) pairs and 10 safe pairs, but with boat side the full state space has 32 combinations, 20 of them safe.
Initial state	(3,3,L).	All six people and boat begin on the left.
Actions	Crossing loads: 1M, 2M, 1C, 2C, 1M+1C; move the load from the boat's current bank to the other bank.	Only loads of one or two people are allowed; boat cannot cross empty.
Goal test	M=0 and C=0 (equivalently everyone is on the right and boat has crossed).	All people must reach the right bank.
Legality check	On each bank, missionaries must be zero or at least as numerous as cannibals: (M=0 or M≥C) and (3−M=0 or 3−M≥3−C).	Both banks must satisfy the safety condition.
Path cost	1 per boat crossing; total path cost is number of crossings.	The objective is the fewest crossings.
Note N1	Goal formulation → problem formulation → search → execution.	This is the standard problem-solving-agent sequence.

(d) Apply your corrected formulation.

(i) Complete the table of boat loads available from the initial state, writing each resulting state as (M, C, boat side) and whether it is legal, with the reason.

Boat load	Resulting state (M, C, boat)	Legal? Reason
1M	(2,3,R)	Illegal: left bank has 2M and 3C, so cannibals outnumber missionaries.
2M	(1,3,R)	Illegal: left bank has 1M and 3C.
1C	(3,2,R)	Legal: left has 3M,2C; right has 0M,1C (zero missionaries is safe).
2C	(3,1,R)	Legal: left has 3M,1C; right has 0M,2C (zero missionaries is safe).

1M + 1C	(2,2,R)	Legal: left has 2M,2C; right has 1M,1C.

(ii) Give one state that the student's legality check accepts although it is actually illegal.

(iii) Using your corrected path cost, state the cost of an optimal solution.

(ii)	Counterexample: (M,C)=(1,0) passes the left-bank check, but the right bank has 2M and 3C, so it is unsafe.
(iii)	11 crossings, with cost 11 because each crossing costs 1. A standard optimal sequence is: CC→, C←, CC→, C←, MM→, MC←, MM→, C←, CC→, C←, CC→.

Question 3: Searching for Solutions and Uninformed Search – Error Analysis of Search Traces

Topic: Searching for solutions, BFS, DFS, uniform-cost search

A student ran BFS, DFS and uniform-cost search (UCS) on the weighted undirected graph below. The start node is S and the goal node is G. **Conventions:** graph search with an explored set; neighbours expanded alphabetically; goal test when a node leaves the frontier; ties broken alphabetically. BFS and DFS ignore edge costs while searching, but path costs are measured with them. **BFS:** do not add a node that is already explored or already in the queue. **DFS:** use a stack with its top on the left; push children so that the alphabetically first child is popped first; do not push a node that is already explored or already on the stack. **UCS:** write entries as Node:g(parent) and replace an entry whenever a cheaper path to a frontier node is found.

Edge	Cost	Edge	Cost
S – C	3	B – G	3
S – D	3	D – E	2
A – B	2	D – G	6
A – C	6	E – G	3
B – D	4		

Student's BFS trace (queue initially [S])

Iteration	Node Dequeued	Queue after Expansion	Explored Set
1	S	[C, D]	{S}
2	C	[D, A]	{S, C}
3	D	[B, E, G, A]	{S, C, D}
4	B	[E, G, A]	{S, C, D, B}
5	E	[G, A]	{S, C, D, B, E}
6	G (goal)	[A]	{S, C, D, B, E}

BFS result: path S – D – G, cost 9, 6 nodes removed from the frontier.

Student's DFS trace (stack initially [S], top on the left)

Iteration	Node Popped	Stack after Pushing Children
1	S	[D, C]
2	D	[G, E, B, C]
3	G (goal)	[E, B, C]

DFS result: path S – D – G, cost 9, 3 nodes removed from the frontier.

Student's UCS trace

Iteration	Node Expanded	g(n)	Priority Queue after Expansion

1	S	0	C:3(S), D:3(S)
2	C	3	D:3(S), A:9(C)
3	D	3	E:5(D), B:7(D), A:9(C), G:9(D)
4	E	5	B:7(D), A:9(C), G:9(D)
5	B	7	A:9(C), G:9(D)
6	A	9	G:9(D)
7	G (goal)	9	–

UCS result: path S – D – G, cost 9, 7 nodes removed from the frontier.

Student's remarks

No.	Remark
R1	An explored set is not needed in BFS because BFS never revisits a node, so the Explored Set column is unnecessary.
R2	All three strategies returned S – D – G with cost 9, so all three are optimal and the traces are confirmed correct.
R3	DFS removed the fewest nodes from the frontier (3), so DFS is the best strategy for this graph.

(a) Examine the three traces and the remarks. List every error, giving the trace and iteration (or the remark number) and what is wrong. An error that is only carried forward counts once.

No.	Where (trace / iteration / remark)	What is wrong
1	BFS iter. 3	D expansion should not enqueue B,E,G ahead of already queued A; alphabetical queue order must preserve A first.
2	BFS iter. 6	Explored set omits G when it is removed for goal test; the goal is removed from frontier then tested.
3	BFS result	S–D–G is a shortest-depth path, but cost is 3+6=9; it is not optimal by cost compared with UCS cost 8.
4	DFS iter. 1	Children order is reversed: alphabetical-first child must be popped first, so stack should be [C,D], not [D,C].
5	DFS trace/result	Correct DFS goes S,C,A,B,G (goal tested on pop), not S,D,G.
6	UCS iters. 3–7	D–G cost is 9, but E gives G cost 8; UCS must replace frontier G:9(D) with G:8(E). It should stop at G cost 8.
7	R1–R3	R1 is false: graph search needs explored/frontier duplicate checks. R2 wrongly equates equal paths/costs with optimality; R3 wrongly calls DFS best based only on fewest removals.

(b) Root cause analysis: for each error listed in (a), explain the rule or concept that was violated and how it led to the wrong entry or conclusion.

No.	Root cause (rule or concept violated)
1	Duplicate detection is required in graph search because different routes can reach the same node; explored and frontier membership prevent repeated expansion.
2	Goal testing is specified on removal from frontier, and the removed goal should be included in explored/processed record.

3	BFS minimizes number of edges, not weighted path cost; S–D–G costs 9 while S–D–E–G costs 8.
4	Stack top is left and alphabetical first must be popped first, so push/order must produce [C,D].
5	DFS follows the corrected stack order and the given goal-test convention; the student trace follows the reversed stack.
6	UCS orders by cumulative cost and replaces a frontier path whenever a cheaper path is found: G via E costs 3+2+3=8.
7	An explored set is needed; optimality differs by strategy and DFS node count alone does not make it best.

(c) Write the corrected traces and justify each correction.

(i) BFS: the corrected queue after iteration 3 and the full order in which nodes are dequeued.

Queue after iteration 3	[A, B, E, G]
Order of dequeued nodes	S, C, D, A, B, E, G

(ii) DFS: the corrected trace.

Iteration	Node Popped	Stack after Pushing Children
1	S	[C, D]
2	C	[A, D]
3	A	[B, D]
4	B	[G, D]
5	G (goal)	[D]

(iii) UCS: the corrected trace.

Iteration	Node Expanded	g(n)	Priority Queue after Expansion
1	S	0	C:3(S), D:3(S)
2	C	3	D:3(S), A:9(C)
3	D	3	E:5(D), B:7(D), A:9(C), G:9(D)
4	E	5	B:7(D), G:8(E), A:9(C)
5	B	7	G:8(E), A:9(C)
6	G (goal)	8	A:9(C)

Justification of the corrections	BFS: A was already in the queue when D expanded, so retain it at the front and enqueue only new nodes. DFS: stack order must make the alphabetically first child pop first. UCS: E→G has cost 8, improving the previous G cost 9; goal is removed at cost 8.

(d) Apply the corrected traces.

(i) Complete the comparison table.

Strategy	Path Found	Path Cost	Nodes Removed from Frontier	Optimal Here?
BFS	S–D–G	9	6	No (not least cost; UCS finds cost 8)
DFS	S–C–A–B–G	14	5	No (DFS is not cost-optimal; path is depth 4)
UCS	S–D–E–G	8	6	Yes (least cumulative cost)

(ii) Use iteration 3 of the corrected BFS trace to explain why an explored set is needed in graph search.

(ii)	At iteration 3, D connects to B, E and G, while A is already in the frontier. The queue/explored checks prevent A or other already-discovered nodes from being added repeatedly. This avoids duplicate frontier entries and repeated expansions.

Question 4: Search with Partial Information – Error Analysis of Belief-State Computation

Topic: Problem formulation, search with partial information (sensorless and percept-based belief-state search)

A vacuum agent lives in a two-square world (Left square, Right square), each square being Clean (C) or Dirty (D). Actions: Left, Right and Suck (Left in the left square and Right in the right square leave the agent where it is; Suck cleans the current square). The goal is that both squares are Clean. Physical states are written (Agent, Left square, Right square):

State	Description	State	Description
S1	(L, D, D)	S5	(L, C, D)
S2	(R, D, D)	S6	(R, C, D)
S3	(L, D, C)	S7	(L, C, C)
S4	(R, D, C)	S8	(R, C, C)

Task A – sensorless agent. The agent knows it is in the Left square but nothing about the dirt, so its initial belief state is {S1, S3, S5, S7}. A student applied the plan [Suck, Right, Suck]:

Step	Action	Belief State after Action
0	(start)	{S1, S3, S5, S7}
1	Suck	{S5, S7}
2	Right	{S6, S8}
3	Suck	{S6, S8}

Task B – agent with a local sensor that reports [location, status of the current square]. Starting from the belief state of all eight states, the agent receives the percept [Left, Dirty], performs Suck, receives [Left, Clean], performs Right and receives [Right, Dirty]. The student's table:

Step	Action	After Action (Predicted)	Percept	After Percept (Updated)
0	(start)	{S1, S2, S3, S4, S5, S6, S7, S8}	[Left, Dirty]	{S1, S2, S3, S6}
1	Suck	{S4, S5, S7, S8}	[Left, Clean]	{S5, S7}
2	Right	{S6, S7}	[Right, Dirty]	{S6}

Student's conclusions

No.	Conclusion
C1	Task A: the final belief state still contains two states, so the agent can never know where it is. No plan can guarantee the goal when the agent has no sensors; hence sensorless problems cannot be solved.
C2	Task B: the belief state is now the singleton {S6}, so the agent knows its location and the dirt status, and the goal has been reached.

C3	With 8 physical states there are only 8 belief states (one per physical state), so searching over belief states is no harder than searching over physical states.

(a) Examine the tables and the conclusions. List every error, giving the task and step (or the conclusion number) and what is wrong. An error that is only carried forward from an earlier wrong row counts once.

No.	Where (task / step / conclusion)	What is wrong
1	Task A, step 3	Suck at Right must clean the right square: S6→S8 and S8→S8, so final belief is {S8}, not {S6,S8}.
2	Task B, step 0	Percept [Left, Dirty] filters all states to those with agent Left and left square Dirty: {S1,S3}, not {S1,S2,S3,S6}.
3	Task B, step 1 predicted	From {S1,S3}, Suck leads to {S5,S7}; the student's {S4,S5,S7,S8} incorrectly includes states outside the current belief.
4	Task B, step 2 predicted	Right from {S5,S7} leads to {S6,S8}, not {S6,S7}.
5	C1	A two-state belief does not mean the location is unknowable forever; actions can transform uncertainty and a sensorless plan can still guarantee a goal.
6	C2	{S6} identifies R and right Dirty, but left is Clean and right is Dirty, so the goal (both clean) has not been reached.
7	C3	With 8 physical states, there are 2^8=256 subsets/belief states, not 8.

(b) Root cause analysis: for each error listed in (a), explain the concept or rule of belief-state search that was misapplied.

No.	Root cause (concept or rule misapplied)
1	Belief-state transitions apply the action to every state in the current set, then take the union of results.
2	Percept update retains only states consistent with every reported percept component.
3	Actions must be applied only to states in the current belief; belief sets do not spontaneously expand to unrelated states.
4	Right moves every state from Left to Right while preserving dirt status, so S7 becomes S8.
5	Belief uncertainty can be reduced or safely managed through a plan; solution means every possible resulting state satisfies the goal.
6	A singleton belief gives full state knowledge, but goal achievement must be checked separately against both squares being clean.
7	A belief state is any subset of physical states, so n physical states yield 2^n possible beliefs.

(c) Write the corrected tables and conclusions, with justification.

(i) Task A: the corrected belief states.

Step	Action	Belief State after Action
0	(start)	{S5, S7}
1	Suck	{S6, S8}

2	Right	{S8}
3	Suck	

(ii) Task B: the corrected belief states after each action and each percept.

Step	Action	After Action (Predicted)	Percept	After Percept (Updated)
0	(start)	{S1, S2, S3, S4, S5, S6, S7, S8}	[Left, Dirty]	{S1, S3}
1	Suck	{S5, S7}	[Left, Clean]	{S5, S7}
2	Right	{S6, S8}	[Right, Dirty]	{S6}

(iii) Corrected conclusions C1 and C2, with justification.

C1	C1 corrected: The final Task A belief is {S8}, which is a singleton state (R,C,C), so the plan guarantees both squares clean. Sensorless problems can be solved when a plan reaches the goal from every state in the initial belief.
C2	C2 corrected: {S6} means the agent is at Right, with left Clean and right Dirty. It knows the physical state, but the goal has not been reached because the right square is dirty.

(d) Apply the corrected concepts.

(i) How many different belief states are possible when there are 8 physical states? Why does this make belief-state search harder than search over physical states?

(i)	256 belief states, because every subset of the 8 physical states is a possible belief state (2^8). This is larger than 8 physical states, so belief-state search may explore many more nodes.

(ii) For the initial belief state {S1, S3, S5, S7}, apply the plan [Right, Suck], write the belief state after each action, and state whether it is a solution.

Step	Action	Belief State after Action
0	(start)	{S1, S3, S5, S7}
1	Right	{S2, S4, S6, S8}
2	Suck	{S4, S8}
	Solution? Why?	No. S4=(R,D,C) has the left square dirty, so not every state in the final belief satisfies the goal.

Rubrics for Error Analysis

Criteria	Weightage (%)	Level 4 – Exemplary (4)	Level 3 – Proficient (3)	Level 2 – Developing (2)	Level 1 – Beginning (1)
Error Identification	**20**	Accurately identifies all errors with clear understanding of issues	Identifies most errors with minor omissions	Identifies limited errors; some are missed	Fails to identify key errors
Root Cause Analysis	**30**	Clearly explains underlying causes with strong technical reasoning	Explains causes with moderate clarity	Partial explanation; weak reasoning	Unable to identify causes of errors
Correction & Justification	**25**	Provides correct solutions with strong justification and logic	Provides correct corrections with some justification	Corrections are partially correct or weakly justified	Incorrect or no corrections provided
Application of Concepts	**15**	Effectively applies relevant concepts to analyze and resolve errors	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts
Clarity & Presentation	**10**	Presents analysis clearly, logically, and systematically	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Q. No. / Criteria	Error Identification	Root Cause Analysis	Correction & Justification	Application of Concepts	Clarity & Presentation	Sub. Total	Total %
1	5	7.5	6.25	3.75	2.5	25	100%
2	5	7.5	6.25	3.75	2.5	25	
3	5	7.5	6.25	3.75	2.5	25	
4	5	7.5	6.25	3.75	2.5	25	

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____